Table 12.6 **Number of electrons in subshell**

Shell number (Symbol)	1 (K)	2 (L)		3 (M)			4 (N)			
Sub shell	1s	2s	2p	3s	3p	3d	4s	4p	4d	4f
Maximum number of electrons in each sub shell	2	2	6	2	6	10	2	6	10	14
Maximum number of electrons in each shell	2	8		18			32			

In the periodic table, it is placed at the top of the alkali metals.

(i) Hydrogen can lose its only one electron to form a hydrogen ion (H^+) like alkali metals.

(ii) It can also gain one electron to form the hydride ion (H^-) like halogens.

(iii) Alkali metals are solids while hydrogen is a gas.

Hence the position of hydrogen in the modern periodic table is still under debate as the properties of hydrogen are unique.

12.2.6. Position of Rare Gases

The elements Helium, Neon, Argon, Krypton, Xenon and Radon of group 18 in the periodic table are called as Noble gases or Rare gases. They are monoatomic gases and do not react with other substances easily, due to completely filled subshells. Hence they are called as inert gases. They are found in very small quantities and hence they are called as rare gases.

12.3 Metals, Non-Metals and Metalloids

12.3.1 Metals

Metals are typically hard, shiny, malleable (can be made as sheet), fusible and ductile (can be drawn into wire) with good electrical and thermal conductivity. Except mercury, most of the metals are solids at room temperature. Metals occupy larger area in the periodic table and are categorized as:

(i) Alkali metals. e.g. Lithium to Francium (top to bottom)

(ii) Alkaline earth metals. e.g: Beryllium to Radium (top to bottom)

(iii) Transition Metals. Group 3 to 12

(iv) p-block metals. e.g: Al, Ga, In, Tl, Sn, Pb and Bi.

4.3.2. Non-metals

A non-metal is an element that does not have the characters like hardness, shiny, malleable, suitable and ductile. In other words, a non-metal is an element that does not have the properties of metal. All non metals are arranged in p-block only. E.g. C, N, O, P, S, Se, Halogen (F, Cl, Br and I) and inert gases (He to Rn).

12.3.3 Metalloids

Elements which have the properties of both metals and non-metals are called as metalloids. (eg) Boron, Arsenic.

12.4 Alloys



During 3500 BC(BCE), people used an alloy named 'bronze'. The idea of making an alloy was quite old. The majority of the metallic substances used today are alloys. Alloys are mixtures of two or more metals and are formed by mixing molten metals thoroughly. Rarely nonmetals are also mixed with metals to produce alloys.

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